

Designing a Vertical-Axis Wind Turbine While Evaluating an AI-Assisted Engineering Workflow

Interim Technical Report Draft

Reporting period: 22 June 2026 through 3 August 2026

Project status: Active; approximately two weeks remain.

Prepared by: Anna and Julie

Review status: Draft for revision. Replace bracketed placeholders, select approved figures, and confirm external-facing language before submission.

Abstract

This project has two linked objectives: develop a vertical-axis wind turbine concept for a Boston Logan International Airport context and evaluate how AI tools can support and accelerate an engineering learning and design workflow. The work began with VAWT and fluid-mechanics research, review of a prior HRI project report, and construction of a source-grounded large-language-model wiki. The wiki was designed to preserve source traceability, identify uncertainty, and support evaluation of AI responses rather than accepting fluent output as correct.

The research corpus supported a screening of 63 VAWT designs. After location-specific criteria, independent human scoring, AI-assisted weighted rankings, and sensitivity tests using alternative weighting personas, the team selected two concepts for further exploration: the EN0005 self-start Darrieus concept and the dual-H-rotor hybrid. The original concept ranking placed Darrieus first at 4.40/5 and hybrid second at 4.25/5 under startup- and efficiency-heavy weights; different expert-performance weights reversed that narrow ordering.

AI-assisted CAD tools produced early geometry assemblies very quickly, but later inspection revealed important limitations of the tools. The Darrieus VAWT lacks sufficient published geometry for a full reconstruction and had some unconnected blade-end bodies. The Hybrid is more fully specified but generated models showed disconnected components and, in one Fusion result, an incorrect inner-blade airfoil. Neither model was structurally adequate after just one prompt and supporting pictures, but after some back and fourth, and manual editing the team got the hybrid model CFD-ready.

The CFD work began with NACA0018 airfoil validation attempt at Reynolds number 50,000 and 5 degrees angle of attack which was ultimately unsuccessful due to time constraints. A pseudo-two-dimensional, rounded-trailing-edge case predicted approximately $Cl = 0.57$ and $Cd = 0.028-0.030$; which remains below the $Cl = 0.75$ AirfoilTools comparison. A NACA0012 control using the same key setup produced $Cl = 0.614$ and $Cd = 0.028$, but it can not be

used not validate the NACA0018 case. With more time the team would have continued to refine the mesh, and in particular would have focused on the boundary layers. The appropriate conclusion at this stage is that the result is a model-specific prediction requiring a condition-matched benchmark, not a validated aerodynamic coefficient for the final turbine.

1. Introduction

1.1 Project purpose

The project was framed as a an exercise vertical axis wind turbine design and as an investigation of practical AI use in engineering. The team did not set out to make an AI driven design of a wind turbine to demonstrate that an AI system could autonomously design a turbine. Instead, the project examined whether a deliberately constrained system could help novice engineers find and organize information, identify knowledge gaps, and learn technical concepts while retaining human control over consequential decisions.

This distinction shaped the work from the outset. The team recognized early on the ways that AI can fail, and built safeguards into the project structure to avoid these pitfalls. The project by tracked four main risks of AI usage: fluency mistaken for correctness, automation bias, anchoring on an initial answer, and sycophancy. The LLM wiki was therefore built with source-only answer rules, citation requirements, explicit labels for unsupported material, and an evaluation loop intended to expose failures.

1.2 Scope and reporting boundary

This report covers work completed during an eight week span. It includes research and wiki construction, site and concept selection, CAD experimentation, and the current CFD-validation effort. It does not claim a final VAWT design, a validated turbine power curve, regulatory approval for installation, a completed CFD validation, structural analysis, manufacturing readiness, or measured turbine performance.

2. Research and Knowledge-System Stage

2.1 Establishing the technical baseline

During the first week, Anna and Julie reviewed fluid-mechanics concepts, a previous capstone report, VAWT literature, and report appendices while meeting with the HRI 99P team. This initial study established the technical vocabulary and identified the need to learn both about VAWT design and the new software workflow before attempting to begin the design phase.

The team also set up OpenCode, Visual Studios, Obsidian, Slack, GitHub, and VS Code Live Share. The early collaboration work included an introductory LLM-wiki workshop, repository creation, shared GitHub access, and asynchronous Git practice. These setup tasks were

mandatory project work that help set up the baseline knowledge of the structure of the wiki. The resulting repository was intended to be the shared technical record for research, decisions, CAD, CFD, AI-use lessons, and to organize all the information gathered about VAWTs.

2.2 Building a source-grounded LLM wiki

The team organized the repository so that raw source material was separated from the editable wiki. Broad behavior and quality rules were retained in `AGENTS.md`, while detailed conversion, ingestion, extraction, organization, and maintenance procedures were separated into `schema/` documents. This separation was introduced after early failures made it clear that general instructions did not reliably produce all required steps.

The source-ingestion workflow converted research PDFs to Markdown while attempting to preserve headings, figures, captions, equations, source identifiers, and traceability. Python and PyMuPDF were added to improve image extraction. The team subsequently repaired incorrectly converted files, restored original PDFs that had been removed before the current procedure, corrected unwanted source summarization, repaired image placement and links, and added missing metadata.

The wiki was organized around concepts, methods, designs, and parameters rather than one undifferentiated note per paper. This architecture enabled comparison of design choices such as airfoil, blade count, startup behavior, rated speed, cut-in speed, and power coefficient. Obsidian tags, properties, backlinks, bases, graphs, and Dataview were explored to make relationships among sources and decisions visible.

2.3 Reliability findings from the research stage

The wiki became useful for source-grounded retrieval and quiz-based learning, but it did not reliably perform every ingestion task without supervision. Recorded failures included timeouts, missing figures and equations, poor formatting, malformed links, altered source content, incomplete metadata, and difficulty extracting graph-derived values such as C_p . The team responded by tightening procedures and manually checking outputs rather than treating the generated pages as automatically correct. The team discovered that using different models lead to variable outputs which were hard to manage. Through creating specific repeatable tasks with instructions pages, like on how to ingest sources or extract figures, these failures significantly decreased in frequency.

This stage established a central project result: information architecture and verification determine whether AI-assisted retrieval is useful. A polished answer, a complete-looking wiki page, or a calculated ranking is not independently valid unless its underlying source evidence and assumptions are checked.

3. Site Definition and Design Requirements

3.1 Boston Logan context

The team narrowed the open-ended VAWT question to Boston Logan International Airport, focusing on Governor's Island as a preliminary location. Logan was selected because high-resolution wind data were accessible and the location offered a plausible high-energy-demand context with stated decarbonization interest. The report record describes an average wind speed of approximately 5 m/s ; the concept-ranking record uses an average wind speed of 4.8 m/s and average gusts of 6.3 m/s . These values should be reconciled against the underlying wind-data analysis before final submission.

Two years of site wind data were added to the wiki and used to produce a wind-speed histogram and guide early design priorities. The team also examined airport airspace regulations and identified Governor's Island as a preliminary candidate area. The $\geq 700 \text{ ft}$ runway-edge criterion is a planning note in the design-goal record, not evidence of a confirmed airport installation approval.

3.2 Design criteria

The design goal prioritized competitive efficiency, low-speed operation appropriate to the site, manufacturability, minimal maintenance, economic feasibility, airport-regulation compliance, CAD feasibility, and compatibility with good CFD practices within the project timeline.

For concept screening, the team translated these priorities into seven criteria: startup ability with a target of $\leq 3 \text{ m/s}$, efficiency/economics, rated-speed fit around $10\text{--}15 \text{ m/s}$, cut-out-speed fit around $\geq 20 \text{ m/s}$, room for future iteration, CAD/manufacturing feasibility, and team interest. Startup and efficiency each received 35% of the original weighted score, followed by CAD/manufacturability at 15%, rated-speed fit at 10%, and room for growth and cut-out fit at 2.5% each.

Uncertainty: The target power, final installation point, site-specific height limit, generator/load curve, and approved airport clearance constraints remain open. The current criteria are suitable for early screening, not a final requirements specification.

4. Concept Selection

4.1 From 63 designs to a focused candidate set

The research wiki contained 63 VAWT designs when the team began formal screening. Anna and Julie conducted a first pass, with LLM assistance, and identified approximately 8-10 promising concepts each; after comparison, they selected six designs for detailed scoring. The candidates represented straight-bladed H-rotor Darrieus, helical Darrieus, reduced-interference hybrid, Savonius-derived, involute, shifted-troposkien, and other source-specific concepts.

The process deliberately combined independent human scoring with AI-assisted ranking. The LLM was used to retrieve structured evidence, compile comparison tables, and calculate weighted scores. Human scores were retained for qualitative criteria such as team interest and perceived CAD difficulty. The team then varied the criterion weights for beginner, expert, and performance-focused personas to determine whether a narrow result depended on one assumption set.

4.2 Ranking result

Under the initial weighting, the self-starting straight-bladed H-rotor Darrieus path ranked first with a weighted score of 4.40/5, and VJ20 ranked second at 4.25/5. In the narrower H-type comparison, VA9 ranked first at 4.30/5 and VJ20 second at 4.25/5. The difference was small enough that the team advanced both concepts instead of treating a single calculated score as a final selection.

The two selected concepts represented a meaningful tradeoff. VA9 offered strong reported low-speed startup evidence, including a reported 1.25 m/s self-start and $C_p = 0.416$; VJ20 offered a higher reported C_p near 0.486, a reported full-scale cut-in of 2.81 m/s, and a dual-rotor architecture with more complexity. These values are source-specific design-screening inputs, not predictions for a future Logan installation.

The sensitivity analysis reinforced the decision to keep both designs active. VA9 remained higher under startup-, simplicity-, and beginner-focused weights, while VJ20 ranked higher in the expert BOS-performance scenario, scoring 4.28/5 compared with VA9's 4.13/5.

4.3 A quantitative-AI failure that changed the process

Manual review later found that the LLM had misreported VJ20's cut-in speed during the ranking process. Julie's log records that the original model output cited a 2.9 m/s value from another design, while a later check of a secondary source reported VJ20's cut-in at 1.5 m/s; the current worksheet lists the VJ20 full-scale cut-in as 2.81 m/s. The conflicting values demonstrate that the exact cut-in metric, scale, and source context must be resolved from the primary VJ20 paper before it is used in a final ranking.

The important outcome was not that an AI-derived number was corrected once. It was that the team changed its operating rule: quantitative values, especially those missing from text or presented in figures, require primary-source verification before they influence a consequential design choice.

5. CAD Development

5.1 Objective and tools

CAD was used as a feasibility test and a preparation stage for later CFD, not as proof of a final turbine design. The team explored Zoo Design Studio and its ZooKeeper agent, including KCL, and later compared Fusion's Autodesk Assistant. The tools were evaluated for their ability to convert research-supported specifications into editable geometry and to support changes without losing traceability.

The team recorded CAD prompts and distinguished source-backed dimensions from inferred support hardware. This record was necessary because a CAD tool can create convincing geometry even when the source does not specify every structural dimension, airfoil coordinate, or connection detail.

5.2 VA9 reconstruction

VA9 was difficult to reconstruct faithfully because the paper did not provide complete CAD-ready geometry. The documented source-backed dimensions were a 36.0 cm blade-body height, 48.0 cm rotor height, 17.3 cm rotor radius, and 5.3 cm chord. The source described a main blade body with two inward-positioned blade ends, but it did not provide a coordinate set for the EN0005 profile, every support dimension, or a definitive blade-end angle.

The team created a parameterized VA9 assembly and iterated primarily on the blade ends. The final logged approximation retained a 46 deg inward rake and smoother lofted transitions after testing rotated extrusion, trim, loft behavior, intermediate profiles, cup depth, outer-lip width, and a denser profile trace. These are modeling choices for an approximation, not source-validated aerodynamic dimensions.

The VA9 model retains a major limitation: the body and blade ends could not be robustly boolean-unioned after the profile became denser, so they remain touching but separate bodies. Therefore, a visually continuous render should not be interpreted as a watertight solid suitable for manufacture or CFD.

5.3 VJ20 reconstruction

VJ20 provided a more complete published rotor specification. The model used two concentric straight-bladed H-rotors on a shared shaft: three outer NACA0018 blades, three inner DU 06-W-200 asymmetric-airfoil blades, 120 deg spacing within each three-blade rotor, and a 60 deg phase offset between the rotors. The prompts prohibited unreported features such as a Savonius rotor, helicity, taper, variable pitch, shrouds, and guide vanes; support hardware was intentionally labeled as inferred and editable.

ZooKeeper generated complex initial assemblies rapidly, but tool-level success did not guarantee usable geometry. The team observed disconnected blades, floating blocks around inner blades, a missing inner top strut after a repair, incorrect inner brackets, and instability when changing rotor azimuth. Reading KCL and asking the agent to identify modified code lines

improved the team's ability to inspect and repair the model, but did not eliminate the need for independent CAD expertise.

Fusion generated full-scale and $1/11.01$ scaled VJ20 models more quickly than ZooKeeper in one trial. Later inspection found that the Fusion inner blades did not use the required DU 06-W-200 asymmetric airfoil. The model is therefore a useful visual and CFD-planning starting point, not a source-faithful VJ20 reproduction.

5.4 CAD result

The CAD work showed that AI can accelerate first-pass parametric geometry, but it also made visual inspection, source traceability, and manual model understanding more important. VA9 remains useful as a parameterized approximation with unresolved source geometry and multi-body issues. VJ20 remains the more tractable baseline because more source geometry is available, but its inner-airfoil fidelity, connections, watertightness, and CFD-specific simplification still require verification.

6. CFD Setup and Airfoil Validation

6.1 Purpose of the validation study

Before using CFD to compare VAWT designs, the team attempted to establish a defensible airfoil workflow. The active study uses a NACA0018 airfoil at $Re = 50,000$, 5° angle of attack, 1.0 m chord, inlet velocity 0.7645 m/s , and air density 1.196 kg/m^3 . The early comparison target was an AirfoilTools curve, approximately $C_L = 0.75$ at the stated condition.

The common later setup was an incompressible, steady-state $k-\omega$ SST case with a velocity inlet, pressure outlet, slip outer boundaries, no-slip airfoil walls, and lift and drag coefficients defined along $+y$ and $+x$, respectively. The study was repeatedly revised as the team learned about finite-span effects, pseudo-two-dimensional modeling, coefficient reference area, boundary layers, y^+ , mesh quality, non-orthogonality, and trailing-edge geometry.

6.2 Key experimental sequence

The first batch used several external-flow domains and mesh choices. NACA0018 predictions ranged from $C_L = 0.1306$ to 0.261 and $C_d = 0.0397$ to 0.061 , substantially below the lift comparison value. One early pseudo-two-dimensional run produced no usable results because result controls had not been configured. These outcomes were exploratory and did not provide a validation basis.

The team then identified a comparability problem: an initial finite-span, three-dimensional airfoil case cannot be directly compared with two-dimensional sectional data. Reducing the spanwise flow volume to 0.5 m and correcting the reference area from 1 m^2 to 0.5 m^2 changed the recorded NACA0018 result from $C_L = 0.291$, $C_d = 0.026$ with the inconsistent area to $C_L =$

0.583 , $C_d = 0.052$ with the corrected area. The change illustrates that coefficient normalization and dimensionality are part of the physical setup, not postprocessing details.

Mesh and near-wall tests then altered mesh fineness, first-layer thickness, wall treatment, turbulence intensity, non-orthogonal correctors, and the trailing-edge geometry. Increasing fineness to 8 produced $C_L = 0.515$, $C_d = 0.041$. Rounding the trailing edge, using a 0.00001 m first layer, and applying two non-orthogonal correctors produced a lower maximum y^+ of 4.913 with $C_L = 0.570$ and $C_d = 0.028$; increasing fineness to 7 produced $C_L = 0.572$ and $C_d = 0.030$.

6.3 Control cases and interpretation

Using the same key rounded-tail setup, the NACA0012 control produced $C_L = 0.614$, $C_d = 0.028$, and maximum $y^+ = 5.33$. Direct STEP inspection confirmed approximately 1 m chord, nominal 12% and 18% thicknesses for the NACA0012 and NACA0018 geometries, and a 0.5 m² reference area for the retained pseudo-two-dimensional wetted section.

The NACA0012 control supports the conclusion that no currently observed shared domain, reference-area, orientation, scale, or rounded-tail-CAD issue explains the NACA0018 discrepancy. It does not prove the NACA0018 AirfoilTools comparison is incorrect, validate the rounded NACA0018 as an ideal sharp profile, establish mesh independence, or validate a VAWT CFD workflow.

The literature record also identifies a low-turbulence NACA0018 experiment with different low-Reynolds-number behavior, including $C_{Lmax} = 0.435$ at $Re = 50,000$ and 3 deg . This means AirfoilTools is not the only relevant reference context, and the remaining discrepancy cannot be assigned confidently to one solver setting.

6.4 Current CFD status

The best documented NACA0018 result is approximately $C_L = 0.57$ and $C_d = 0.028-0.030$ for a steady, $k-\omega$ SST , rounded-NACA0018 pseudo-two-dimensional SimScale case. It should be reported exactly as that model-specific prediction, not as an experimentally validated coefficient.

Later mesh refinement work improved mesh appearance and varied boundary-layer count, overall thickness, growth rate, surface sizing, volume refinements, and local refinement placement. Results remained inconsistent, and the current record includes residual, non-orthogonality, and reference-benchmark concerns. The team has appropriately shifted from attempting to force agreement with one target toward documenting the case and seeking a condition-matched benchmark and expert review.

7. Assessment of AI in the Workflow

7.1 High-value uses

The strongest observed AI uses were retrieval across a curated source set, organization of research into comparable notes, generation of candidate work plans, calculation of weighted decision scores, initial CAD generation, and guided learning through explanations and quizzes. These uses accelerated the movement from a broad literature collection to a structured design conversation.

The design-selection workflow demonstrates the benefit concretely. AI made repeated ranking and sensitivity scenarios practical, while human-defined criteria and independent scores preserved a way to inspect the result. CAD tools similarly made it possible to produce and compare complex initial assemblies before the team could have modeled them conventionally at the same level of detail.

7.2 Limits and failures

The weakest observed uses were unsupervised quantitative extraction, reliable source conversion and formatting, tracking the full state of repeated CFD iterations, and autonomous technical troubleshooting. The VJ20 cut-in-speed misreport, malformed conversions, missing figures, wrong CAD airfoil, disconnected geometry, and repetitive CFD advice are evidence that AI output must be treated as a proposal for review rather than a completed engineering task.

The project also surfaced a learning tradeoff. Fast synthesis can reduce the time spent engaging directly with papers and struggling through technical concepts, which may reduce deep understanding. The team responded by using the wiki as a teaching tool, asking it to quiz and explain concepts, reviewing source documents, and preserving records of assumptions and failures.

7.3 Interim workflow principle

The evidence supports a human-led workflow: use AI to broaden search, organize information, expose alternatives, perform transparent calculations, and create first-pass artifacts; use humans to define the objective, judge source quality, verify numbers, inspect geometry, choose acceptable tradeoffs, and decide when a result is sufficiently validated.

8. Conclusions and Remaining Work

By 3 August, the project had completed a substantial research and screening phase, built a source-grounded technical knowledge system, selected VA9 and VJ20 as leading concepts, generated exploratory CAD models, and created a detailed record of an incomplete airfoil CFD-validation effort. The work has not yet produced a validated final VAWT, but it has established a traceable foundation for the remaining design and validation tasks.

The next two weeks should prioritize the smallest set of actions that produce defensible evidence: verify the baseline CAD geometry and its CFD suitability, resolve the CFD benchmark and mesh-quality acceptance criteria, run only staged simulations whose controls and outputs are confirmed, and document the final AI-workflow lessons from both successes and failures.

Uncertain: The final VAWT geometry, airport installation feasibility, full-turbine CFD results, aerodynamic performance, and manufacturing feasibility are not established by the current record.

Check next: Confirm source-level VJ20 operating values; select a matched NACA0018 benchmark; verify CAD watertightness, airfoil identity, and face groups; define CFD mesh, convergence, timestep, and periodicity acceptance criteria before interpreting a turbine result.

Appendix A. Project Timeline Through 3 August 2026

Period	Major completed work	Evidence
Week 1, 22-26 June	Reviewed fluid mechanics, previous report, VAWT research, and project references; set up OpenCode, VS Code, Obsidian, Slack, GitHub, and Live Share; began LLM-wiki workshop, source ingestion, evals, and shared repository workflow.	active/_anna daily log.md; active/_Julie_daily_log.md
Week 2, 29 June-3 July	Repaired source conversions; added Python/PyMuPDF-assisted image extraction; created schema/ procedures; improved metadata, links, figures, equations, wiki organization, design pages, and parameter pages.	active/_anna daily log.md; active/_Julie_daily_log.md
Week 3, 6-10 July	Evaluated Boston wind data and Logan siting; created design goal and work plan; screened and ranked designs; selected VA9 and VJ20 for further study; began Zoo/ZooKeeper CAD exploration.	active/_anna daily log.md; active/_Julie_daily_log.md; active/analysis/Concept ranking worksheet.md

Period	Major completed work	Evidence
Week 4, 13-17 July	Learned SimScale and CFD fundamentals; began NACA0018 validation; added SimScale and targeted CFD sources; explored pseudo-two-dimensional setup, reference area, mesh refinement, y^+ , and control cases.	active/_anna daily log.md; active/_Julie_daily_log.md; active/documentation/CFD log/Airfoil Validation Studies.md
Week 5, 20-24 July	Continued airfoil validation and mesh troubleshooting; created full-scale and scaled VJ20 CAD; attempted VJ20 CFD setup; identified the VJ20 cut-in reporting issue; met with Ryan and Philip for workflow and CFD guidance.	active/_anna daily log.md; active/_Julie_daily_log.md
Week 6, 27-31 July	Continued mesh refinement with expert feedback; installed COMSOL as a possible fallback; inspected geometry defects affecting mesh quality; began full-turbine setup while continuing reporting and documentation.	active/_anna daily log.md; active/_Julie_daily_log.md
Week 7, 3 August	Continued SimScale work after geometry cleanup and added documentation images; two remaining weeks were still planned at the reporting cutoff.	active/_Julie_daily_log.md

Appendix B. Concept-Selection Evidence Snapshot

Item	Recorded value or outcome	Interpretation limit
Design corpus	63 VAWT designs in the wiki	Count supports scope of screening, not equal evidence quality for every design.
Detailed concepts	6 designs after the initial human/LLM pass	The initial shortlist contained human judgment and may not be exhaustive.
Initial ranking weights	Startup 35% ; efficiency 35% ; CAD/manufacturability 15% ; rated-speed fit 10% ; growth 2.5% ; cut-out fit 2.5%	Weights represent project priorities, not universal VAWT optimization weights.
VA9 result	4.40/5 in original weighted table; reported 1.25 m/s self-start; reported $C_p = 0.416$	Source-specific evidence; not a Logan performance prediction.

Item	Recorded value or outcome	Interpretation limit
VJ20 result	4.25/5 in original weighted table; reported $C_p = 0.486$; reported 2.81 m/s full-scale cut-in	Cut-in records conflict across project notes and require primary-source reconciliation.
Sensitivity result	VA9 stronger under practicality/startup weightings; VJ20 stronger under expert-performance weighting (4.28/5 vs 4.13/5)	Supports advancing both concepts, not declaring one universal winner.

Appendix C. CAD Geometry Ledger

Concept	Source-backed geometry carried into CAD	Inferred or unresolved geometry	Current limitation
VA9 EN0005 self-start Darrieus	36.0 cm blade-body height; 48.0 cm rotor height; 17.3 cm rotor radius; 5.3 cm chord; main body plus inward blade ends	EN0005 coordinates, support dimensions, definitive blade-end angle, and full structural design	Blade body and ends remain separate touching bodies after boolean-union failure; model is an approximation.
VJ20 dual H-rotor hybrid	Three outer NACA0018 blades; three inner DU 06-W-200 blades; 120 deg blade spacing; 60 deg rotor phase offset	Hubs, struts, bearings, generator mount, and other mechanical details	Zoo models showed disconnected/floating components; Fusion inner blades did not match the required asymmetric airfoil.

Appendix D. Airfoil CFD Validation Record

D.1 Case definition

Parameter	Value
Airfoil	NACA0018
Angle of attack	5 deg
Reynolds number	50,000
Chord	1.0 m
Inlet velocity	0.7645 m/s

Parameter	Value
Air density	1.196 kg/m ³
Primary model	Incompressible, steady-state k- ω SST
Pseudo-2D reference area	0.5 m ² for 1.0 m chord and 0.5 m wetted span

D.2 Selected results

Case	Key change	Cl	Cd	Note
Batch 1 Runs 1-5	Early domains and meshes	0.1306– 0.261	0.0397– 0.061	Exploratory cases; inadequate validation basis.
Batch 2 Run 5	Pseudo-2D span and corrected 0.5 m ² area	0.583	0.052	Area correction materially changed reported coefficient.
Run 8	Mesh fineness 8	0.515	0.041	Improved drag but lower lift.
Run 13	Rounded tail, near-wall changes, two non-orthogonal correctors	0.570	0.028	Maximum $y^+ = 4.913$; best documented low-drag case.
Run 14	Run 13 with fineness 7	0.572	0.030	Result effectively unchanged.
Run 15 control	NACA0012 with Run 13 settings	0.614	0.028	Supports the shared workflow but does not validate NACA0018.

D.3 CFD limitations to retain in final report

- No complete mesh-independence, domain-independence, convergence, transition-model, or cycle-repeatability study has been documented.
- The rounded NACA0018 geometry is not an ideal sharp trailing-edge NACA0018 and cannot be treated as an exact match to every external reference.
- AirfoilTools and the low-Reynolds-number c_{j9} experiment represent different reference contexts; neither alone closes the validation question.
- The NACA0010 supplementary drag value is recorded as 0.27 while also described as correct; it should be checked against the saved SimScale output before inclusion in any final report.

Appendix E. Recommended Figures and Evidence to Insert

1. A project workflow diagram: research sources -> conversion and procedures -> wiki -> site/design criteria -> ranking -> CAD -> CFD -> human verification. Use a newly created diagram rather than implying automation alone made decisions.
2. A Logan wind-speed histogram with its data period, station, units, and processing method stated in the caption. Do not use an unlabeled screenshot.
3. The concept ranking table with weights and explicit uncertainty labels for inferred or missing data.
4. A VA9 render beside an annotation of source-backed versus inferred geometry.
5. A VJ20 render with blade-count, rotor-phase, and airfoil-verification status identified.
6. A CFD mesh and coefficient-history figure that identifies the exact run, geometry, mesh settings, reference area, and convergence status.

Appendix F. Source Records Used for This Draft

- active/_anna daily log.md
- active/_Julie_daily_log.md
- active/analysis/Design goal.md
- active/analysis/Decision Making Process.md
- active/analysis/Concept ranking worksheet.md
- active/documentation/Anna blog + report writing.md
- active/documentation/CAD Modeling Process Blog Post.md
- active/documentation/CFD log/Airfoil Validation Studies.md
- active/documentation/CFD log/Airfoil Validation Study Summary and Report Outline.md
- active/documentation/Weeks 1-4 Project Summary and Report Outline.md
- sources/cj9.md